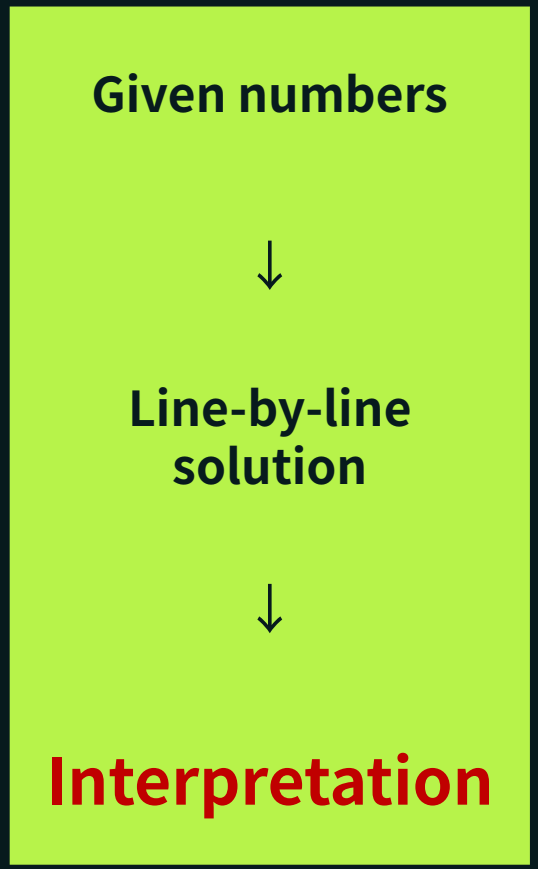
Time-Series Causality & Dynamics Hand-Calculation Workbook

Seven problems solved step by step — no calculator needed



Financial Time Series · 08 Workbook

All seven problems are solved in the same three steps

From stationarity to cointegration — every calculation in the seven main decks, redone by hand

01
Stationarity
Half-life 3.1

02
ADL
LR multiplier 1.0

03
Granger
F = 4.0

04
VAR
Eigenvalues 0.6, 0.3

05
IRF · FEVD
Reorder: 25%

06
SVAR
PP′ = S

07
Cointegration
Half-life 1.9

3 steps per problem
① Given numbers
② Line-by-line
③ Interpretation

※ The number at the bottom of each box is that problem's final answer — cover it before you start

A single logarithm solves for the half-life

Basics — if a shock shrinks by a factor b each period, only b to the power k remains after k periods

① Definition → solve

$$b^k = 0.5 \;\Rightarrow\; k \ln b = \ln 0.5 \;\Rightarrow\; k = \frac{\ln 0.5}{\ln b}$$

② Handy log values

$$\ln 0.5 = -0.693 \quad \ln 0.7 = -0.357 \quad \ln 0.8 = -0.223$$
$$\ln 0.9 = -0.105 \quad \ln 0.95 = -0.0513 \quad \ln 1 = 0$$

③ When b is near 1

$$k \approx \frac{0.693}{1-b} \qquad (b \to 1)$$

Why logs: the unknown k sits in the exponent, so a log brings it down

ln 0.5 = −0.693 is the numerator of every half-life in this workbook

Four 2×2 matrix operations are all you need

Basics — product, transpose, trace and determinant, and the eigenvalue formula

① Product

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} u \\ w \end{pmatrix} = \begin{pmatrix} pu + qw \\ ru + sw \end{pmatrix}$$

② Transpose

$$P = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix}, \quad P' = \begin{pmatrix} p_{11} & p_{21} \\ 0 & p_{22} \end{pmatrix}$$

③ Trace · det

$$\mathrm{tr} = p + s, \qquad \det = ps - qr$$

④ Eigenvalues

$$\text{eigenvalue} = \frac{\mathrm{tr} \pm \sqrt{\mathrm{tr}^2 - 4\det}}{2}$$

The VAR, IRF and SVAR problems just repeat these four operations

01

Stationarity and Shock Half-Life

The larger the AR(1) coefficient b, the longer a shock persists — three cases: b = 0.8, 0.95, 1

We find how long an AR(1) shock takes to halve

Problem 1 — a one-time shock of 1 unit, with no new shocks afterwards

Model

$$y_t = b\, y_{t-1} + e_t$$

Fraction left

$$1 \;\to\; b \;\to\; b^2 \;\to\; \cdots \;\to\; b^k$$

Given

$$b = 0.8, \qquad b = 0.95, \qquad b = 1$$

Find: the half-life k in each case — the time until only half the shock remains

b = 0.8 takes 3.1 periods; b = 0.95 takes 13.5 periods

Step-by-step — plug the log values into the half-life formula

① Definition

$$b^k = 0.5$$

② Solve for k

$$k = \ln 0.5 \,/\, \ln b$$

③ b = 0.8

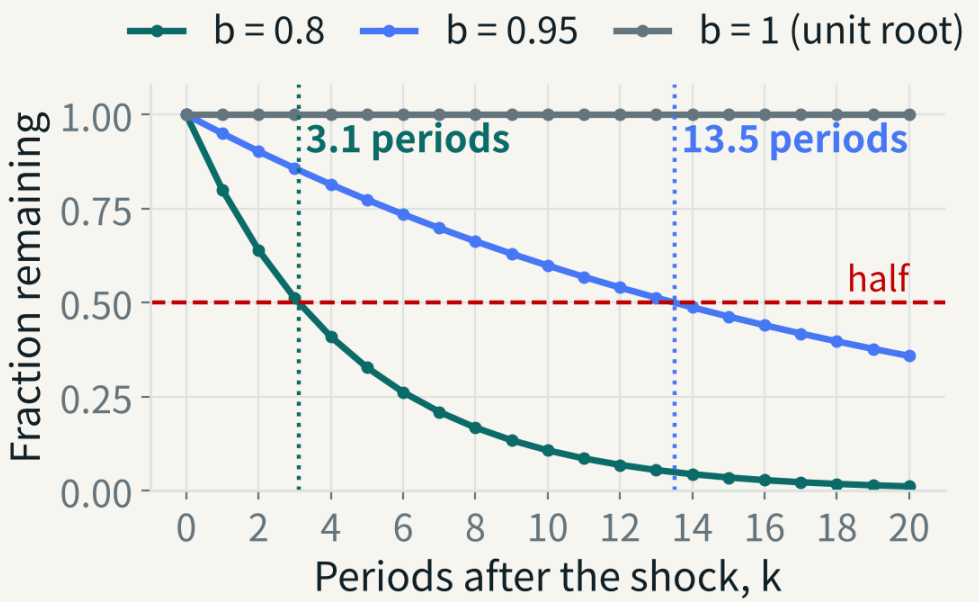
$$k = -0.693 \,/\, -0.223 \approx 3.1$$

④ b = 0.95

$$k = -0.693 \,/\, -0.0513 \approx 13.5$$

⑤ b = 1

$$\ln 1 = 0 \;\Rightarrow\; k = \infty$$



Takeaway: as b nears 1 the half-life explodes, making it hard to tell from a unit root

02

ADL Dynamic and Long-Run Multipliers

How a one-time rise in x spreads into y over successive periods

We trace how a one-time shock to x spreads into y

Problem 2 — ADL model; x rises by 1 unit at time t only, then returns to 0

Model

$$y_t = 1 + 0.5\,y_{t-1} + 0.3\,x_{t-1} + 0.2\,x_{t-2} + e_t$$

Coefficients

$$a = 1, \quad b_1 = 0.5, \quad c_1 = 0.3, \quad c_2 = 0.2$$

Path of x

$$x: \ \ldots, 0, \underset{t}{1}, 0, 0, \ldots$$

Find: the dynamic multipliers from t to t+4, and the long-run multiplier (two ways)

The dynamic multipliers run 0, 0.30, 0.35, 0.175, 0.0875

Step-by-step — half of last period's response plus that period's x coefficient

① Recursion $d_h = 0.5\, d_{h-1} + c_h, \qquad c_1 = 0.3,\ c_2 = 0.2,\ \text{else } 0$

② Time t $d_0 = 0 \qquad (\text{no } x_t \text{ term})$

③ t+1 $d_1 = 0.5(0) + 0.3 = 0.30$

④ t+2 $d_2 = 0.5(0.30) + 0.2 = 0.35$

⑤ t+3 $d_3 = 0.5(0.35) = 0.175$

⑥ t+4 $d_4 = 0.5(0.175) = 0.0875$

Takeaway: the response peaks at 0.35 after two periods, then halves each period

Both methods give a long-run multiplier of 1.0

Step-by-step — compare the sum of all responses with the formula

① Running sum

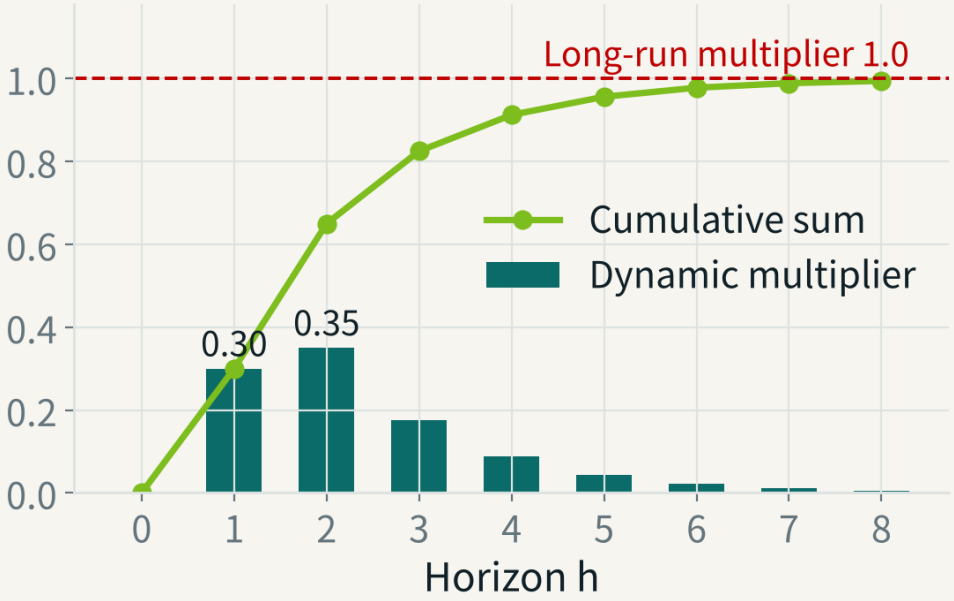
$$k = 0.30 + 0.35 + 0.175 + 0.0875 + \cdots$$

② Geometric sum

$$= 0.30 + \frac{0.35}{1-0.5} = 0.30 + 0.70 = 1.0$$

③ Formula

$$k = \frac{0.3+0.2}{1-0.5} = \frac{0.5}{0.5} = 1.0$$



Takeaway: a permanent unit rise in x raises y by 1.0 in the long run — 100% pass-through

03

Granger Causality F-Test

The gap in residual sums of squares between restricted and unrestricted models decides predictive value

We test whether two lags of x help predict y

Problem 3 — illustrative numbers: residual sums of squares set for practice

F-statistic

$$F = \frac{(RSS_R - RSS_U)/q}{RSS_U/(n-k)}$$

Illustrative

$$RSS_R = 103, \quad RSS_U = 95$$
$$q = 2, \quad n = 100, \quad k = 5$$

Counting q, k

$$k = \underbrace{1}_{a} + \underbrace{2}_{b_1, b_2} + \underbrace{2}_{c_1, c_2} = 5, \qquad q = 2 \;\; (c_1 = c_2 = 0)$$

Find: the F value and how it compares with the 5% critical value

※ Illustrative numbers — RSS and sample size are set for practice, not taken from real data

F = 4.0 exceeds the critical value 3.09, so we reject the null

Step-by-step (illustrative numbers) — compute numerator and denominator, then divide

① Numerator

$$(103 - 95)/2 = 4.0$$

② Denominator

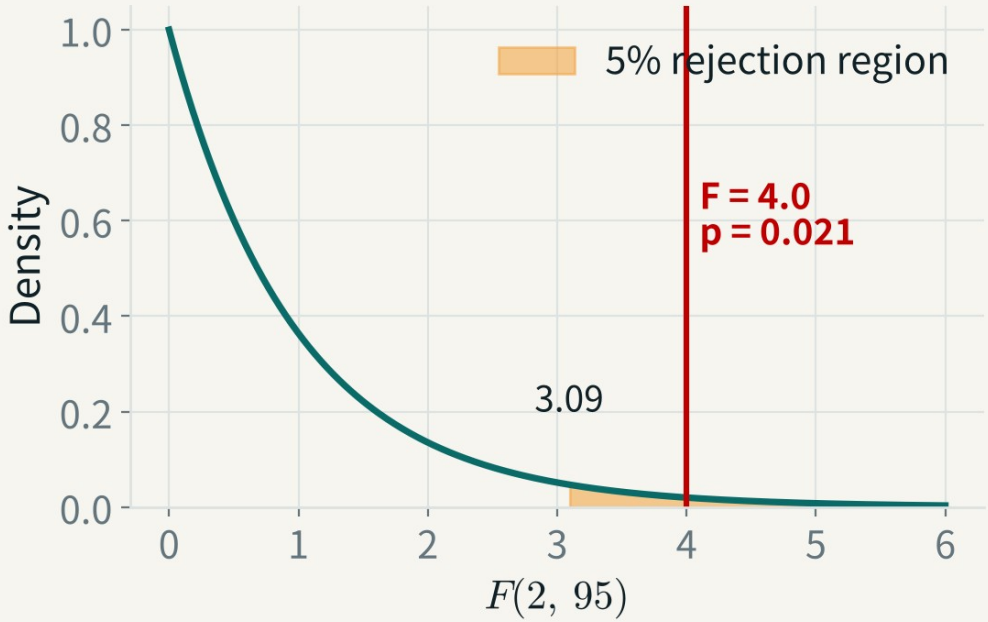
$$95/(100 - 5) = 95/95 = 1.0$$

③ F

$$F = 4.0/1.0 = 4.0$$

④ Compare

$$4.0 > F_{0.05}(2, 95) \approx 3.09$$



Takeaway: past x significantly improves the forecast of y at the 5% level

※ Critical value 3.09 and p ≈ 0.021 are from F(2, 95) via scipy — in practice use a table or software

04

VAR Stability and Impulse Responses

Check stability with the eigenvalues of the coefficient matrix, then roll a shock forward one period at a time

We check VAR(1) stability and trace the response to an x shock

Problem 4 — variable order [y, x]; non-orthogonalized responses that ignore shock correlation

Model

$$z_t = A\, z_{t-1} + e_t, \qquad A = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}$$

1-unit x shock

$$z_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad z_h = A\, z_{h-1}$$

Find: the two eigenvalues of A and stability, then the y and x responses at h = 1–3

Eigenvalues 0.6 and 0.3 are both below 1, so the VAR is stable

Step-by-step — the trace and determinant are all you need for the eigenvalues

① Trace

$$\mathrm{tr} = 0.5 + 0.4 = 0.9$$

② Determinant

$$\det = 0.5(0.4) - 0.2(0.1) = 0.20 - 0.02 = 0.18$$

③ Discriminant

$$\mathrm{tr}^2 - 4\det = 0.81 - 0.72 = 0.09$$

④ Eigenvalues

$$\text{eigenvalue} = (0.9 \pm \sqrt{0.09})/2 = (0.9 \pm 0.3)/2 = 0.6,\ 0.3$$

Takeaway: the larger eigenvalue, 0.6, governs how fast shocks die out

The y response peaks at 0.2 after one period and then declines

Step-by-step — multiply the previous period's vector by A to get the next

① h = 1

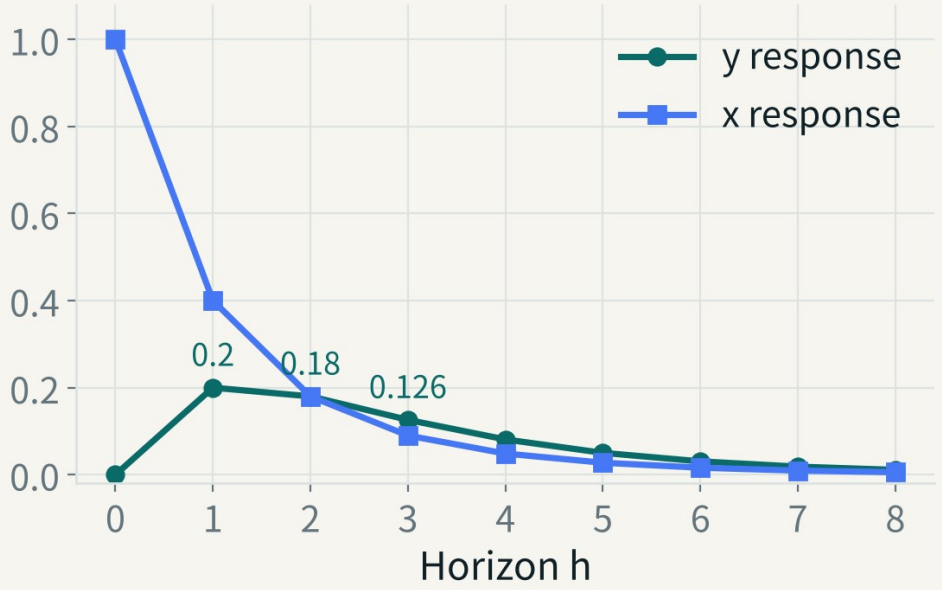
$$z_1 = \begin{pmatrix} 0.5(0) + 0.2(1) \\ 0.1(0) + 0.4(1) \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix}$$

② h = 2

$$z_2 = \begin{pmatrix} 0.5(0.2) + 0.2(0.4) \\ 0.1(0.2) + 0.4(0.4) \end{pmatrix} = \begin{pmatrix} 0.18 \\ 0.18 \end{pmatrix}$$

③ h = 3

$$z_3 = \begin{pmatrix} 0.5(0.18) + 0.2(0.18) \\ 0.1(0.18) + 0.4(0.18) \end{pmatrix} = \begin{pmatrix} 0.126 \\ 0.09 \end{pmatrix}$$



Takeaway: the x shock reaches y from the next period; both responses converge to 0

05

IRF and FEVD — Cholesky Orthogonalization

Assign the shock correlation of 0.5 by variable ordering, then split forecast error variance by shock

We orthogonalize correlated shocks to get IRFs and FEVD shares

Problem 5 — the same A as Problem 4 plus shock covariance S; variable order [y, x]

Given matrices

$$A = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

Tools

$$S = PP', \qquad R_h = A^h, \qquad O_h = R_h P$$

Diagonal of S (1, 1): shock variances; off-diagonal 0.5: covariance (= correlation)

Find: Cholesky factor → orthogonalized IRFs → FEVD of y and x → reversed ordering

The Cholesky factor P is found by matching three cells in turn

Step-by-step — expand PP′ and set it equal to S cell by cell

① Expand PP′

$$PP' = \begin{pmatrix} p_{11}^2 & p_{11}p_{21} \\ p_{11}p_{21} & p_{21}^2 + p_{22}^2 \end{pmatrix} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

② First column

$$p_{11}^2 = 1 \Rightarrow p_{11} = 1, \qquad p_{11}p_{21} = 0.5 \Rightarrow p_{21} = 0.5$$

③ Last cell

$$p_{22} = \sqrt{1 - 0.5^2} = \sqrt{0.75} \approx 0.866$$

④ Result

$$P = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.866 \end{pmatrix}$$

Takeaway: the 0.5 correlation is assigned as 'a y shock moves x by 0.5 in the same period'

Orthogonalized responses O are the response matrix times P

Step-by-step — h = 0 is P itself; for h = 1, 2 multiply the response matrix by P

① 2-step matrix

$$R_2 = A^2 = \begin{pmatrix} 0.25 + 0.02 & 0.10 + 0.08 \\ 0.05 + 0.04 & 0.02 + 0.16 \end{pmatrix} = \begin{pmatrix} 0.27 & 0.18 \\ 0.09 & 0.18 \end{pmatrix}$$

② h = 0

$$O_0 = I\,P = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.866 \end{pmatrix}$$

③ h = 1

$$O_1 = AP = \begin{pmatrix} 0.5 + 0.2(0.5) & 0.2(0.866) \\ 0.1 + 0.4(0.5) & 0.4(0.866) \end{pmatrix} = \begin{pmatrix} 0.600 & 0.173 \\ 0.300 & 0.346 \end{pmatrix}$$

④ h = 2

$$O_2 = R_2 P = \begin{pmatrix} 0.27 + 0.18(0.5) & 0.18(0.866) \\ 0.09 + 0.18(0.5) & 0.18(0.866) \end{pmatrix} = \begin{pmatrix} 0.360 & 0.156 \\ 0.180 & 0.156 \end{pmatrix}$$

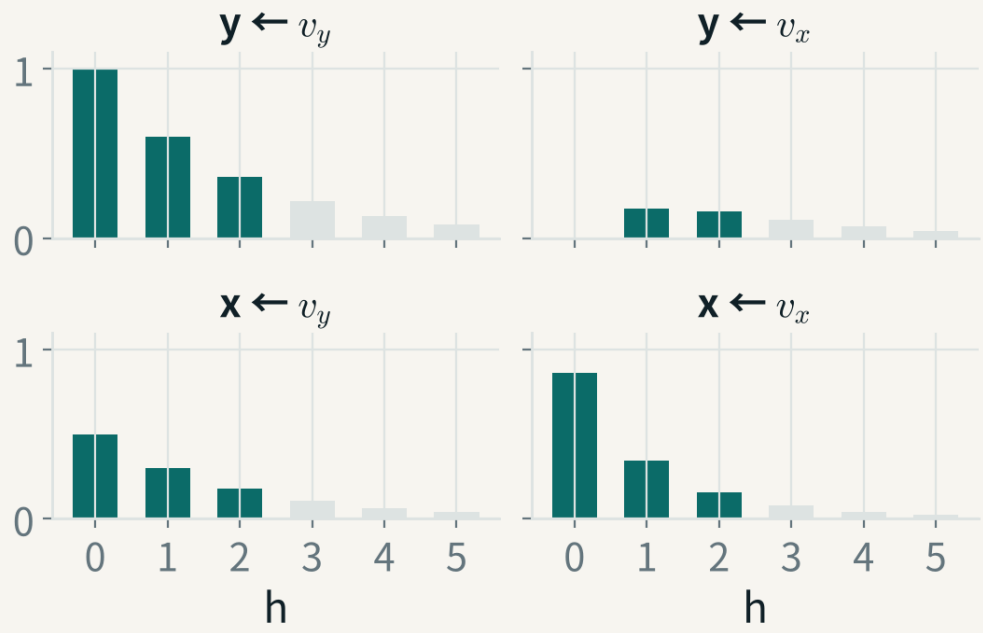
Takeaway: column 1 is the response to a 1-s.d. y shock, column 2 to a 1-s.d. x shock

An x shock cannot move y in the same period

Orthogonalized IRF table — rows are horizons, columns are 'response ← structural shock'

h	y ← y shock	y ← x shock	x ← y shock	x ← x shock
0	**1.000**	**0.000**	**0.500**	**0.866**
1	0.600	0.173	0.300	0.346
2	0.360	0.156	0.180	0.156

※ Row h = 0 equals P (response matrix = I)



Takeaway: ordering y first shows up directly as the 0.000 at h = 0

Through h = 3, over 96% of y's forecast error variance is its own shock

Step-by-step — square and accumulate the responses, then divide by the total

① 1 step ahead

$$v_y:\ 1.000^2 = 1.000, \qquad v_x:\ 0^2 = 0 \ \Rightarrow\ 100\%\,/\,0\%$$

② 2-step cum.

$$v_y:\ 1 + 0.600^2 = 1.360, \qquad v_x:\ 0 + 0.173^2 = 0.030$$

③ 2-step share

$$1.360/1.390 = 97.8\%, \qquad 0.030/1.390 = 2.2\%$$

④ 3-step cum.

$$v_y:\ 1.360 + 0.360^2 = 1.490, \qquad v_x:\ 0.030 + 0.156^2 = 0.054$$

⑤ 3-step share

$$1.490/1.544 = 96.5\%, \qquad 0.054/1.544 = 3.5\%$$

Steps	Total	y shock	x shock
1	1.000	100%	0%
2	1.390	97.8%	2.2%
3	**1.544**	**96.5%**	**3.5%**

Takeaway: y moves almost entirely on its own shock — it behaves as largely exogenous

The y shock explains 25% of x's forecast error variance from the start

Step-by-step — apply the same method to the second row of O (the x responses)

① 1 step ahead

$$v_y:\ 0.500^2 = 0.250, \qquad v_x:\ 0.866^2 = 0.750 \ \Rightarrow\ 25.0\% \,/\, 75.0\%$$

② 2-step cum.

$$v_y:\ 0.25 + 0.300^2 = 0.340, \qquad v_x:\ 0.75 + 0.346^2 = 0.870$$

③ 2-step share

$$0.340/1.210 = 28.1\%, \qquad 0.870/1.210 = 71.9\%$$

④ 3-step cum.

$$v_y:\ 0.340 + 0.180^2 = 0.372, \qquad v_x:\ 0.870 + 0.156^2 = 0.894$$

⑤ 3-step share

$$0.372/1.267 = 29.4\%, \qquad 0.894/1.267 = 70.6\%$$

Steps	Total	y shock	x shock
1	1.000	25.0%	75.0%
2	1.210	28.1%	71.9%
3	**1.267**	**29.4%**	**70.6%**

Check: the 1-step-ahead totals for y and x are both 1.000 — matching the diagonal of S

Reversing the order alone lifts the x-shock share from 0% to 25%

Step-by-step — with x first, the 0.5 correlation is assigned in the x → y direction

① Order [x, y]

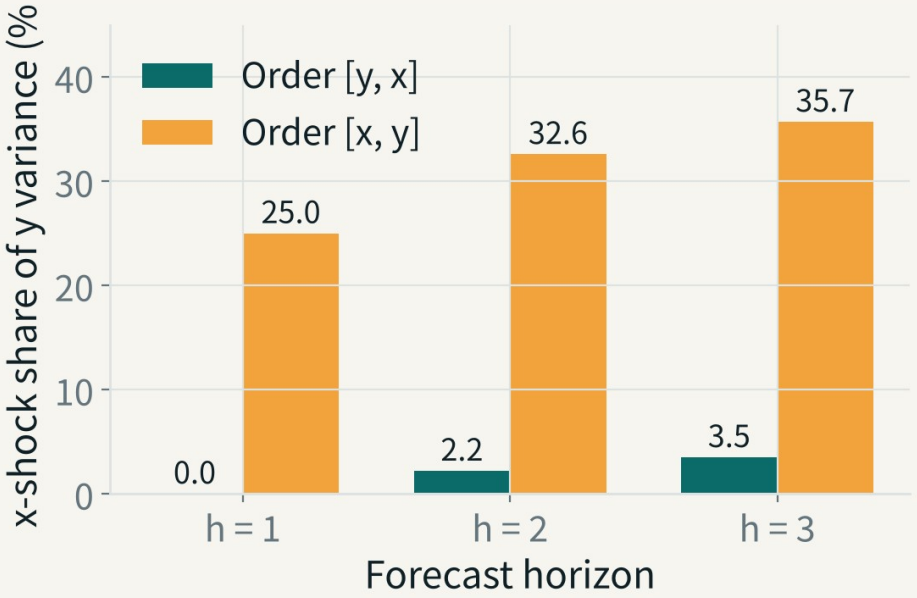
$$\text{order } [x, y]: \quad P = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.866 \end{pmatrix} \quad \text{rows } (x,\ y)$$

② y at h = 0

$$y \leftarrow v_x = 0.5, \qquad y \leftarrow v_y = 0.866$$

③ 1-step share

$$0.5^2/(0.5^2 + 0.866^2) = 0.25/1.00 = 25\%$$



Takeaway: same data, same VAR — the ordering assumption alone changes the conclusion

※ The h = 2, 3 values in the chart (32.6%, 35.7%) were computed in python by the same method

06

SVAR Identification — Same S, Different P

Multiply out to check that two identifying assumptions reproduce the same covariance

We decompose the same covariance S into two structures

Problem 6 — y is output, x is the interest rate; the data reveal only S

Reduced-form cov.

$$S = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

Structural link

$$e_t = P\,v_t, \qquad S = PP'$$

Assumption A · output ordered first

Rate shock → output 0 same month

Assumption B · rate ordered first

Output shock → rate 0 same month

Find: multiply out to confirm that P under both assumptions satisfies PP′ = S

Both assumptions reproduce S exactly

Step-by-step — multiply P by its transpose P′ and compare with the four cells of S

① Assumption A

$$P_A P_A' = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.866 \end{pmatrix} \begin{pmatrix} 1 & 0.5 \\ 0 & 0.866 \end{pmatrix} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 0.25 + 0.75 \end{pmatrix} = S$$

② Assumption B

$$P_B P_B' = \begin{pmatrix} 0.866 & 0.5 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0.866 & 0 \\ 0.5 & 1 \end{pmatrix} = \begin{pmatrix} 0.75 + 0.25 & 0.5 \\ 0.5 & 1 \end{pmatrix} = S$$

Same-month responses under A

Rate→output 0 · output→rate +0.5

Same-month responses under B

Rate→output +0.5 · output→rate 0

Takeaway: the fit is identical; the different conclusions come purely from identification

07

Cointegration — Spread Decay and Half-Life

How much the adjustment speeds close the spread from equilibrium each period

We find how fast the spread from equilibrium closes

Problem 7 — bivariate VECM; short-run dynamics and shocks are ignored

VECM

$$\Delta y_t = a_y\,(y_{t-1} - x_{t-1})$$
$$\Delta x_t = a_x\,(y_{t-1} - x_{t-1})$$

Definition

$$\mathrm{gap}_t = y_t - k\,x_t = y_t - x_t$$

Given

$$k = 1, \quad a_y = -0.2, \quad a_x = 0.1, \quad y - x = 5$$

Find: the spread at t+1 and t+2, the decay factor per period, and the half-life (also b = 0.9)

The spread goes 5 → 3.5 → 2.45, shrinking to 0.7 times each period

Step-by-step — subtract both variables' changes from last period's spread

① Changes, t+1

$$\Delta y = -0.2 \times 5 = -1.0, \qquad \Delta x = 0.1 \times 5 = +0.5$$

② Spread, t+1

$$5 - 1.0 - 0.5 = 3.5$$

③ Changes, t+2

$$\Delta y = -0.2(3.5) = -0.7, \qquad \Delta x = 0.1(3.5) = +0.35$$

④ Spread, t+2

$$3.5 - 0.7 - 0.35 = 2.45$$

⑤ Decay factor

$$\text{gap}_t = (1 + a_y - a_x)\,\text{gap}_{t-1} = 0.7\,\text{gap}_{t-1}$$

Takeaway: y bears 2/3 of the adjustment, x 1/3 — the larger |adjustment speed| moves more

The half-life is 1.9 periods, or 6.6 periods if b = 0.9

Step-by-step — the decay factor 0.7 is the AR(1) coefficient b of the spread

① b = 0.7

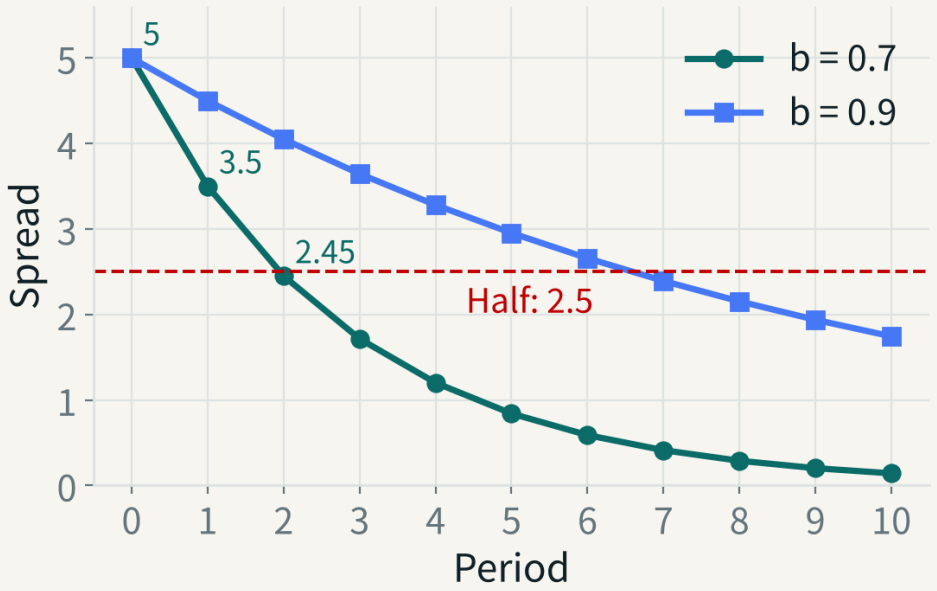
$$k = \ln 0.5 / \ln 0.7 = -0.693 / -0.357 \approx 1.9$$

② b = 0.9

$$k = \ln 0.5 / \ln 0.9 = -0.693 / -0.105 \approx 6.6$$

③ Check

$$0.9^6 = 0.53, \qquad 0.9^7 = 0.48$$



Takeaway: the half-life sizes a pairs-trade holding period — about 7 periods if b = 0.9

Claude Code checks all hand-calculated answers in one pass

Lab — confirm in python that the same numbers come out; if not, find the step where they diverge

Recompute half-lives → **F value & critical value** → **Eigenvalues & Cholesky** → **Rebuild IRF · FEVD** → **Save & open check report**

Prompt to paste (summary)

- Recompute all seven workbook problems with numpy/scipy and tabulate them
- If a value differs from my hand result, show the step where it diverges
- Save the results as workbook_check.md and open it

Students only paste the prompt and check the numbers

Write down the givens,

solve line by line,

interpret in one line.

These seven hand calculations
underpin the main decks.